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java.util.concurrent

Class AbstractExecutorService

java.lang.Object
  java.util.concurrent.AbstractExecutorService

All Implemented Interfaces:

Executor, ExecutorService

Direct Known Subclasses:

ForkJoinPool, ThreadPoolExecutor

```
public abstract class AbstractExecutorService
extends Object
implements ExecutorService
```

Provides default implementations of ExecutorService execution methods. This class implements the submit, invokeAny and invokeAll methods using a RunnableFuture returned by newTaskFor, which defaults to the FutureTask class provided in this package. For example, the implementation of submit(Runnable) creates an associated RunnableFuture that is executed and returned. Subclasses may override the newTaskFor methods to return RunnableFuture implementations other than FutureTask.

Extension example. Here is a sketch of a class that customizes ThreadPoolExecutor to use a CustomTask class instead of the default FutureTask:

```
 public class CustomThreadPoolExecutor extends ThreadPoolExecutor {

   static class CustomTask<V> implements RunnableFuture<V> {...}

   protected <V> RunnableFuture<V> newTaskFor(Callable<V> c) {
       return new CustomTask<V>(c);
   }
   protected <V> RunnableFuture<V> newTaskFor(Runnable r, V v) {
```

```
        return new CustomTask<V>(r, v);
    }
    // ... add constructors, etc.
}
```

Since:

1.5

Constructor Summary

Constructors

Constructor	Description
`AbstractExecutorService()`	

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`<T> List<Future<T>>`	`invokeAll(Collection<? extends Callable<T>> tasks)`	Executes the given tasks, returning a list of Futures holding their status and results when all complete.
`<T> List<Future<T>>`	`invokeAll(Collection<? extends Callable<T>> tasks, long timeout, TimeUnit unit)`	Executes the given tasks, returning a list of Futures holding their status and results when all complete or the timeout expires, whichever happens first.
`<T> T`	`invokeAny(Collection<? extends Callable<T>> tasks)`	Executes the given tasks, returning the result of one that has

		completed successfully (i.e., without throwing an exception), if any do.
<T> T	**invokeAny**(**Collection**<? extends **Callable**<T>> tasks, long timeout, **TimeUnit** unit)	Executes the given tasks, returning the result of one that has completed successfully (i.e., without throwing an exception), if any do before the given timeout elapses.
protected <T> **RunnableFuture**<T>	**newTaskFor**(**Callable**<T> callable)	Returns a RunnableFuture for the given callable task.
protected <T> **RunnableFuture**<T>	**newTaskFor**(**Runnable** runnable, T value)	Returns a RunnableFuture for the given runnable and default value.
<T> **Future**<T>	**submit**(**Callable**<T> task)	Submits a value-returning task for execution and returns a Future representing the pending results of the task.
Future<?>	**submit**(**Runnable** task)	Submits a Runnable task for execution and returns a Future

		representing that task.
<T> **Future**<T>	**submit**(**Runnable** task, T result)	Submits a Runnable task for execution and returns a Future representing that task.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Methods inherited from interface java.util.concurrent.ExecutorService

awaitTermination, isShutdown, isTerminated, shutdown, shutdownNow

Methods inherited from interface java.util.concurrent.Executor

execute

Constructor Detail

AbstractExecutorService

```
public AbstractExecutorService()
```

Method Detail

newTaskFor

```
protected <T> RunnableFuture<T> newTaskFor(Runnable runnable,
                                           T value)
```

Returns a RunnableFuture for the given runnable and default value.

Type Parameters:

T - the type of the given value

Parameters:

`runnable - the runnable task being wrapped`

`value - the default value for the returned future`

Returns:

`a RunnableFuture which, when run, will run the underlying runnable and which, as a Future, will yield the given value as its result and provide for cancellation of the underlying task`

Since:

`1.6`

newTaskFor

```
protected <T> RunnableFuture<T> newTaskFor(Callable<T> callable)
```

Returns a `RunnableFuture` for the given callable task.

Type Parameters:

`T - the type of the callable's result`

Parameters:

`callable - the callable task being wrapped`

Returns:

`a RunnableFuture which, when run, will call the underlying callable and which, as a Future, will yield the callable's result as its result and provide for cancellation of the underlying task`

Since:

`1.6`

submit

```
public Future<?> submit(Runnable task)
```

Description copied from interface: `ExecutorService`

Submits a Runnable task for execution and returns a Future representing that task. The Future's `get` method will return `null` upon *successful* completion.

Specified by:

`submit` in interface `ExecutorService`

Parameters:

`task - the task to submit`

Returns:

`a Future representing pending completion of the task`

Throws:

RejectedExecutionException - if the task cannot be scheduled for execution

NullPointerException - if the task is null

submit

```
public <T> Future<T> submit(Runnable task,
                            T result)
```

Description copied from interface: ExecutorService

Submits a Runnable task for execution and returns a Future representing that task. The Future's get method will return the given result upon successful completion.

Specified by:

submit in interface ExecutorService

Type Parameters:

T - the type of the result

Parameters:

task - the task to submit

result - the result to return

Returns:

a Future representing pending completion of the task

Throws:

RejectedExecutionException - if the task cannot be scheduled for execution

NullPointerException - if the task is null

submit

```
public <T> Future<T> submit(Callable<T> task)
```

Description copied from interface: ExecutorService

Submits a value-returning task for execution and returns a Future representing the pending results of the task. The Future's get method will return the task's result upon successful completion.

If you would like to immediately block waiting for a task, you can use constructions of the form `result = exec.submit(aCallable).get();`

Note: The Executors class includes a set of methods that can convert some other common closure-like objects, for example, PrivilegedAction to Callable form so they can be submitted.

Specified by:

submit in interface ExecutorService

Type Parameters:

T - the type of the task's result

Parameters:

task - the task to submit

Returns:

a Future representing pending completion of the task

Throws:

RejectedExecutionException - if the task cannot be scheduled for execution

NullPointerException - if the task is null

invokeAny

```
public <T> T invokeAny(Collection<? extends Callable<T>> tasks)
                throws InterruptedException,
                       ExecutionException
```

Description copied from interface: ExecutorService

Executes the given tasks, returning the result of one that has completed successfully (i.e., without throwing an exception), if any do. Upon normal or exceptional return, tasks that have not completed are cancelled. The results of this method are undefined if the given collection is modified while this operation is in progress.

Specified by:

invokeAny in interface ExecutorService

Type Parameters:

T - the type of the values returned from the tasks

Parameters:

tasks - the collection of tasks

Returns:

the result returned by one of the tasks

Throws:

InterruptedException - if interrupted while waiting

ExecutionException - if no task successfully completes

invokeAny

```
public <T> T invokeAny(Collection<? extends Callable<T>> tasks,
                       long timeout,
                       TimeUnit unit)
                throws InterruptedException,
                       ExecutionException,
                       TimeoutException
```

Description copied from interface: ExecutorService

Executes the given tasks, returning the result of one that has completed successfully (i.e., without throwing an exception), if any do before the given timeout elapses. Upon normal or exceptional return, tasks that have not completed are cancelled. The results of this method are undefined if the given collection is modified while this operation is in progress.

Specified by:

`invokeAny` in interface `ExecutorService`

Type Parameters:

`T` - the type of the values returned from the tasks

Parameters:

`tasks` - the collection of tasks

`timeout` - the maximum time to wait

`unit` - the time unit of the timeout argument

Returns:

the result returned by one of the tasks

Throws:

`InterruptedException` - if interrupted while waiting

`ExecutionException` - if no task successfully completes

`TimeoutException` - if the given timeout elapses before any task successfully completes

invokeAll

```
public <T> List<Future<T>> invokeAll(Collection<? extends Callable<T>> tasks)
                            throws InterruptedException
```

Description copied from interface: ExecutorService

Executes the given tasks, returning a list of Futures holding their status and results when all complete. `Future.isDone()` is `true` for each element of the returned list. Note that a *completed* task could have terminated either normally or by throwing an exception. The results of this method are

undefined if the given collection is modified while this operation is in progress.

Specified by:

`invokeAll` in interface `ExecutorService`

Type Parameters:

`T` - the type of the values returned from the tasks

Parameters:

`tasks` - the collection of tasks

Returns:

a list of Futures representing the tasks, in the same sequential order as produced by the iterator for the given task list, each of which has completed

Throws:

`InterruptedException` - if interrupted while waiting, in which case unfinished tasks are cancelled

invokeAll

```
public <T> List<Future<T>> invokeAll(Collection<? extends Callable<T>> tasks,
                                     long timeout,
                                     TimeUnit unit)
                              throws InterruptedException
```

Description copied from interface: `ExecutorService`

Executes the given tasks, returning a list of Futures holding their status and results when all complete or the timeout expires, whichever happens first. `Future.isDone()` is `true` for each element of the returned list. Upon return, tasks that have not completed are cancelled. Note that a *completed* task could have terminated either normally or by throwing an exception. The results of this method are undefined if the given collection is modified while this operation is in progress.

Specified by:

`invokeAll` in interface `ExecutorService`

Type Parameters:

`T` - the type of the values returned from the tasks

Parameters:

`tasks` - the collection of tasks

`timeout` - the maximum time to wait

`unit` - the time unit of the timeout argument

Returns:

a list of Futures representing the tasks, in the same sequential order as produced by the iterator for the given task list. If the operation did not time out, each task will have completed. If it did time out, some of these tasks will not have completed.

Throws:

`InterruptedException` - if interrupted while waiting, in which case unfinished tasks are cancelled

Submit a bug or feature

For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.

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